



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

|              |          |              |                                                      |
|--------------|----------|--------------|------------------------------------------------------|
| Class:       | TE-AI&DS | Semester:    | V                                                    |
| Course Code: | 2015111  | Course Name: | Statistics for Machine Learning and Data Science Lab |

|                          |                                                                               |
|--------------------------|-------------------------------------------------------------------------------|
| Name of Student:         | Parth Manoj Gajare                                                            |
| Roll No. :               | 74                                                                            |
| Experiment No.:          | 1                                                                             |
| Title of the Experiment: | Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study) |
| Date of Performance:     |                                                                               |
| Date of Submission:      |                                                                               |

## Evaluation

| Performance Indicator              | Max. Marks | Marks Obtained |
|------------------------------------|------------|----------------|
| Performance                        | 5          |                |
| Understanding                      | 5          |                |
| Journal work and timely submission | 10         |                |
| Total                              | 20         |                |

| Performance Indicator              | Exceed Expectations (EE) | Meet Expectations (ME) | Below Expectations (BE) |
|------------------------------------|--------------------------|------------------------|-------------------------|
| Performance                        | 4-5                      | 2-3                    | 1                       |
| Understanding                      | 4-5                      | 2-3                    | 1                       |
| Journal work and timely submission | 8-10                     | 5-8                    | 1-4                     |

Checked by

Name of Faculty: Signature :

Date :

2015111: Statistics for Machine Learning and Data Science Lab



# Vidyavardhini's College of Engineering and Technology

## Department of Artificial Intelligence & Data Science

---

**Aim:** To perform exploratory statistical analysis of the Pima Indians Diabetes Dataset by examining its structure, variable types, data quality, statistical characteristics, and important patterns.

### Objectives

After completing this experiment, the student will be able to:

1. Apply exploratory data analysis techniques to understand the structure and characteristics of a real-world healthcare dataset.
2. Interpret the statistical characteristics, data quality issues, and important patterns identified during the analysis.

### Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

### Dataset Details

**Dataset:** Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations, 8 input attributes, and one binary outcome variable.



### Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Examine the dataset dimensions, column names, data types, and sample records.
3. Classify the variables as numerical, categorical, or binary.
4. Generate a statistical summary of the dataset.
5. Check for missing or invalid values.
6. Identify potential outliers using suitable statistical techniques and visualizations.
7. Generate appropriate visualizations to understand the data.
8. Record and interpret important observations obtained from the exploratory analysis.

### Description of the Experiment

Exploratory Data Analysis (EDA) is the initial process of understanding a dataset before applying statistical inference or machine learning techniques.

The analysis follows: Raw Dataset, Data Inspection, Variable Identification, Data Quality, Analysis, Statistical Summary, Visualization and Interpretation

The Pima Diabetes Dataset is explored to understand its structure, data quality, variable distributions, and important characteristics before developing predictive models.

### Detailed Description of Statistical techniques:

#### Descriptive Statistics

Statistical measures such as the following are used to summarize numerical variables:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard Deviation

#### Missing Value Analysis

The dataset is examined for missing or invalid values.



For example, zero values in certain medical attributes such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI may require investigation because a zero value may represent missing or invalid data rather than a genuine medical measurement.

### Outlier Analysis

Potentially unusual observations are identified using statistical summaries and visualizations such as boxplots.

An outlier should be investigated before deciding whether it should be removed or retained.

### Data Visualization

Suitable visualizations are generated to understand:

- Variable distributions
- Frequency of diabetes outcomes      Relationships between variables
- Potential outliers Examples include:
- Histograms
- Boxplots
- Bar charts
- Scatter plots

### CODE:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("diabetes.csv")
```

```
print("Shape of Dataset:", df.shape)
```

```
print("\nColumns:")
print(df.columns)
```

```
print("\nData Types:")
print(df.dtypes)
```

```
print("\nFirst 5 Records:")
print(df.head())
```



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

---

```
print("\nNumerical Variables:")
for col in df.columns:
    if df[col].nunique() > 2:
        print("-", col)
```

```
print("\nBinary Variable:")
print("Outcome")
```

```
print("\nStatistical Summary")
print(df.describe())
```

```
print("\nMedian")
print(df.median())
```

```
print("\nVariance")
print(df.var())
```

```
print("\nStandard Deviation")
print(df.std())
```

```
print("\nMissing Values")
print(df.isnull().sum())
```

```
print("\nZero Values in Important Columns")
```

```
columns = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
```

```
for col in columns:
    print(col, ":", (df[col] == 0).sum())
```

```
print("\nPossible Outliers (IQR Method)")
```

```
for col in df.columns[:-1]:
```

```
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
```



# Vidyavardhini's College of Engineering and Technology

## Department of Artificial Intelligence & Data Science

---

```
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df[col] < lower) | (df[col] > upper)]

print(col, ":", len(outliers))

df.hist(figsize=(12,10))
plt.show()

plt.figure(figsize=(12,8))

for i, col in enumerate(df.columns[:-1]):
    plt.subplot(3,3,i+1)
    sns.boxplot(y=df[col])

plt.tight_layout()
plt.show()

sns.countplot(x="Outcome", data=df)
plt.title("Diabetes Outcome")
plt.show()

plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.show()

sns.scatterplot(data=df, x="Glucose", y="BMI", hue="Outcome")
plt.show()
```



### Output:-

Shape of Dataset: (768, 9)

Columns:

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',  
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],  
      dtype='object')
```

Data Types:

```
Pregnancies      int64  
Glucose           int64  
BloodPressure     int64  
SkinThickness     int64  
Insulin           int64  
BMI               float64  
DiabetesPedigreeFunction float64  
Age               int64  
Outcome           int64  
dtype: object
```

First 5 Records:

|   | Pregnancies | Glucose | BloodPressure | ... | DiabetesPedigreeFunction | Age | Outcome |
|---|-------------|---------|---------------|-----|--------------------------|-----|---------|
| 0 | 6           | 148     | 72            | ... | 0.627                    | 50  | 1       |
| 1 | 1           | 85      | 66            | ... | 0.351                    | 31  | 0       |
| 2 | 8           | 183     | 64            | ... | 0.672                    | 32  | 1       |
| 3 | 1           | 89      | 66            | ... | 0.167                    | 21  | 0       |
| 4 | 0           | 137     | 40            | ... | 2.288                    | 33  | 1       |

[5 rows x 9 columns]

Numerical Variables:

- Pregnancies
- Glucose
- BloodPressure

Numerical Variables:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Binary Variable:

Outcome

Statistical Summary

|       | Pregnancies | Glucose    | BloodPressure | ... | DiabetesPedigreeFunction | Age        | Outcome    |
|-------|-------------|------------|---------------|-----|--------------------------|------------|------------|
| count | 768.000000  | 768.000000 | 768.000000    | ... | 768.000000               | 768.000000 | 768.000000 |
| mean  | 3.845052    | 120.894531 | 69.105469     | ... | 0.471876                 | 33.240885  | 0.348958   |
| std   | 3.369578    | 31.972618  | 19.355807     | ... | 0.331329                 | 11.760232  | 0.476951   |
| min   | 0.000000    | 0.000000   | 0.000000      | ... | 0.078000                 | 21.000000  | 0.000000   |
| 25%   | 1.000000    | 99.000000  | 62.000000     | ... | 0.243750                 | 24.000000  | 0.000000   |
| 50%   | 3.000000    | 117.000000 | 72.000000     | ... | 0.372500                 | 29.000000  | 0.000000   |
| 75%   | 6.000000    | 140.250000 | 80.000000     | ... | 0.626250                 | 41.000000  | 1.000000   |
| max   | 17.000000   | 199.000000 | 122.000000    | ... | 2.420000                 | 81.000000  | 1.000000   |

[8 rows x 9 columns]

Median

```
Pregnancies      3.0000  
Glucose           117.0000  
BloodPressure     72.0000  
SkinThickness     23.0000  
Insulin           30.5000  
BMI               32.0000  
DiabetesPedigreeFunction 0.3725  
Age               29.0000
```



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

```
DiabetesPedigreeFunction    0.3725
Age                        29.0000
Outcome                    0.0000
dtype: float64

Variance
Pregnancies                11.354056
Glucose                    1022.248314
BloodPressure              374.647271
SkinThickness              254.473245
Insulin                    13281.180078
BMI                        62.159984
DiabetesPedigreeFunction    0.109779
Age                        138.303046
Outcome                    0.227483
dtype: float64

Standard Deviation
Pregnancies                3.369578
Glucose                    31.972618
BloodPressure              19.355807
SkinThickness              15.952218
Insulin                    115.244002
BMI                        7.884160
DiabetesPedigreeFunction    0.331329
Age                        11.760232
Outcome                    0.476951
dtype: float64

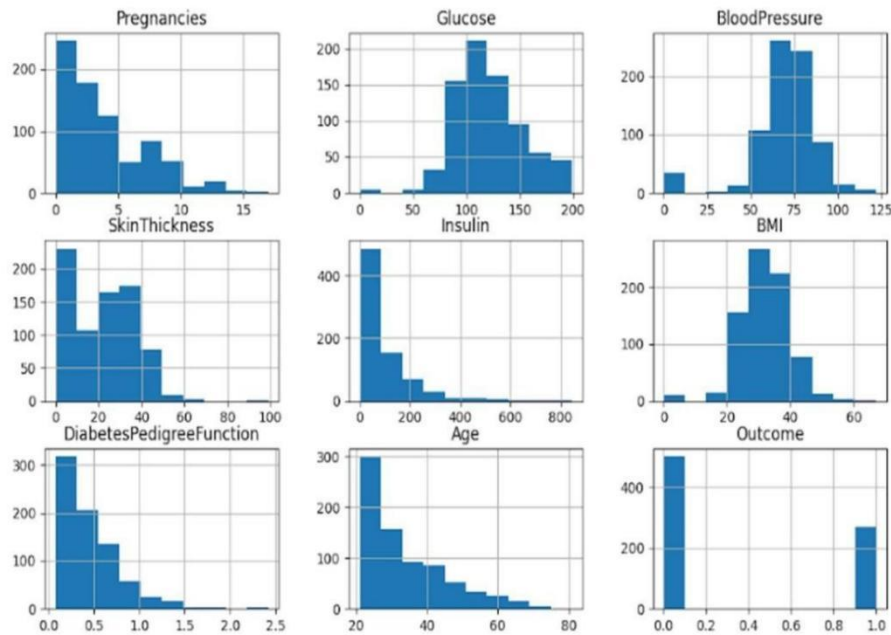
Missing Values
Pregnancies                0
Glucose                    0
BloodPressure              0
SkinThickness              0
Insulin                    0
```

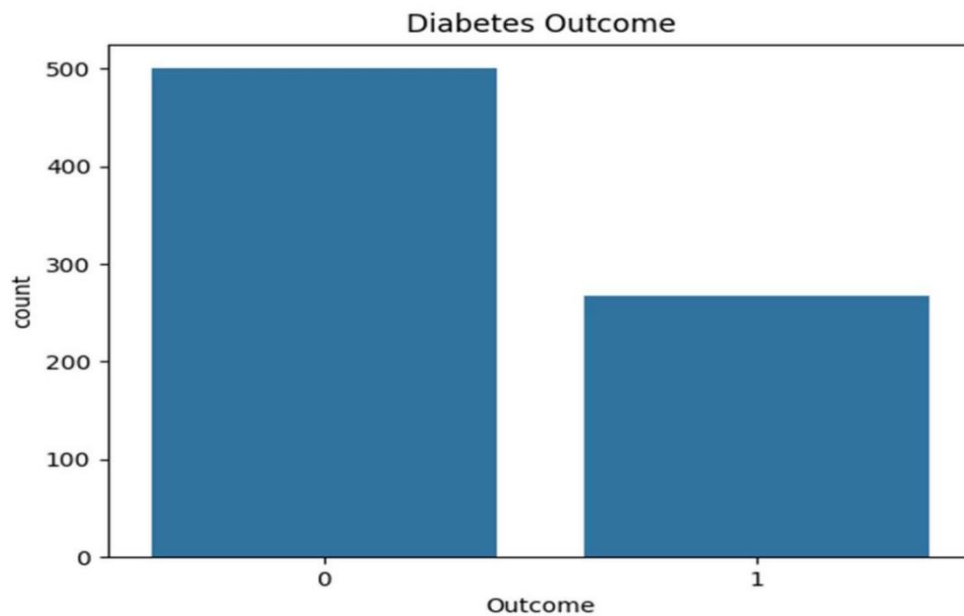
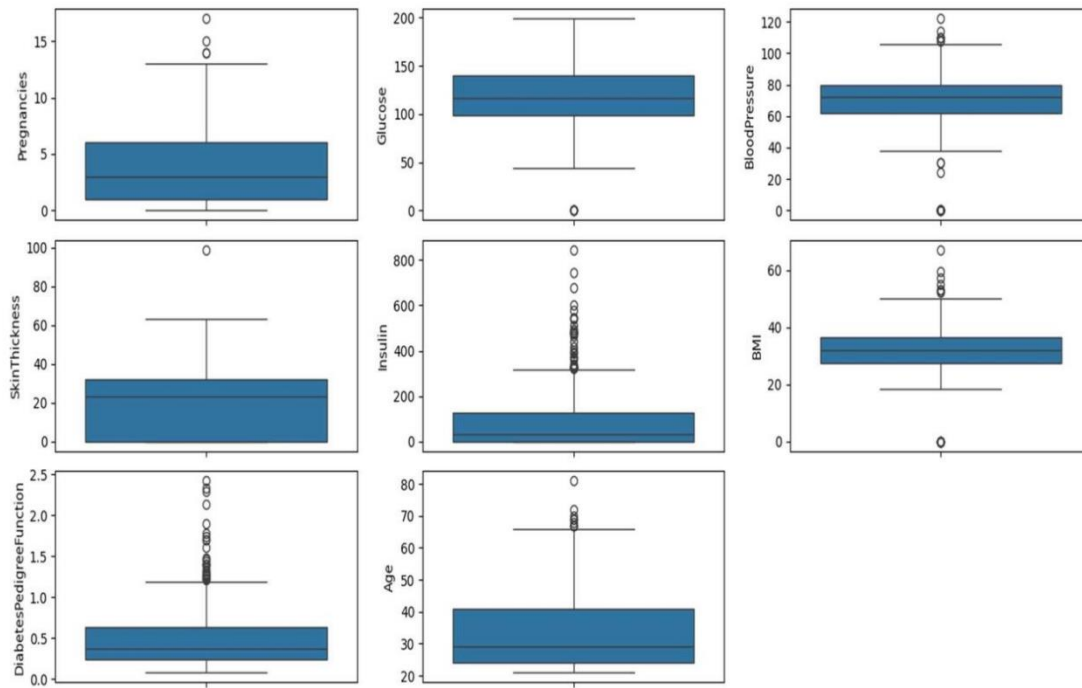
```
Insulin                    115.244002
BMI                        7.884160
DiabetesPedigreeFunction    0.331329
Age                        11.760232
Outcome                    0.476951
dtype: float64

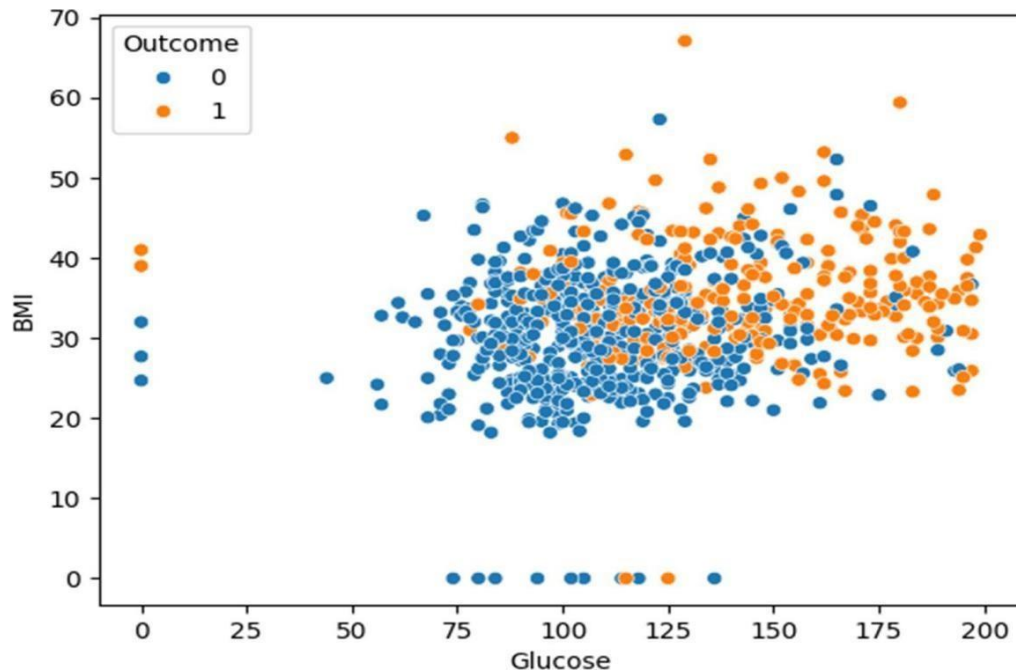
Missing Values
Pregnancies                0
Glucose                    0
BloodPressure              0
SkinThickness              0
Insulin                    0
BMI                        0
DiabetesPedigreeFunction    0
Age                        0
Outcome                    0
dtype: int64

Zero Values in Important Columns
Glucose : 5
BloodPressure : 35
SkinThickness : 227
Insulin : 374
BMI : 11

Possible Outliers (IQR Method)
Pregnancies : 4
Glucose : 5
BloodPressure : 45
SkinThickness : 1
Insulin : 34
BMI : 19
DiabetesPedigreeFunction : 29
Age : 9
```







### Conclusion

Exploratory statistical analysis was performed on the Pima Indians Diabetes Dataset to understand its structure, variable characteristics, data quality, and important patterns. The analysis provides the foundation for further statistical analysis and machine learning model development.

Questions to be Answered by Students

1. Which variables in the dataset contain zero or potentially invalid values that require investigation?

The following variables contain zero values that may represent missing or invalid medical measurements and should be investigated:

- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI



These variables cannot normally have a value of zero in a clinical setting, so such values are often treated as missing data during preprocessing

2. Which variable contains potential outliers based on the analysis?

The variables with potential outliers are:

- Insulin (most significant outliers)
- BMI
- DiabetesPedigreeFunction
- Pregnancies

These outliers were identified using boxplots and the Interquartile Range (IQR) method

3. What are the important patterns observed between the input variables and the Outcome variable?

The important patterns observed are:

- Glucose has the strongest positive relationship with the Outcome variable; diabetic patients generally have higher glucose levels.
- BMI tends to be higher among patients with diabetes.
- Age shows a moderate positive association with diabetes, with older individuals more likely to have diabetes.
- Pregnancies are generally higher among diabetic patients.
- The correlation matrix indicates that Glucose is the most influential predictor, followed by BMI and Age.

4. State two important insights obtained from the exploratory analysis.

1. Glucose is the strongest predictor of diabetes, with diabetic patients having noticeably higher glucose levels than non-diabetic patients.
2. The dataset contains invalid zero values and several outliers, indicating that data preprocessing (handling missing values and evaluating outliers) is necessary before performing statistical analysis or building machine learning models.